

TECHNEXA 2026 • PROBLEM STATEMENT #5

Smart Healthcare Access & Referral

Closing critical gaps between rural patients, frontline clinics, and regional specialists.

— Engineering Brief • Core Problem Gaps & Web Solution

THE CORE BOTTLENECK

The Real-World Referral Breakdown

65%

Paper Referrals Lost in Transit

Underserved patients abandon secondary consultations due to lack of coordination and travel fatigue.

ROOT CAUSES

Three Fatal Gaps in Current Care



Disconnected Data

Zero prior context arrives at hospital.



Facility Mismatch

Patients travel to unavailable specialists.



Zero Feedback Loop

Primary clinics never receive outcome status.

Current Reality vs. Proposed System

The Broken Paper Model

Unchecked: High risk of lost paperwork

Blind Trips: Unknown equipment availability

No Tracking: 65% patient drop-off rate

Our Smart Web Platform

Digital Pass: Single QR code with vital context

Live Matching: Real-time facility capability filter

Closed-Loop: Automated status return to primary clinic

End-to-End Referral Lifecycle

01

Intake

Symptom entry by frontline health worker.

02

Triage

Automated specialty and urgency recommendation.

03

QR Pass

Instant digital voucher for receiving hospital.

04

Close Loop

Consultation logged with automated SMS follow-up.

The Web Application Stack

Frontend Client

Responsive React dashboard with multilingual interface.

Triage Engine

Rule-based specialty mapping and severity sorting.

Hospital Portal

One-click QR scanner with referral status sync.



PROJECTED FEASIBILITY

Target Real-World Outcomes

0%

Lost Referrals

Complete digital record continuity.

3x

Faster Admissions

Instant triage priority on arrival.

45%

Travel Reduction

Zero wasted trips to unequipped sites.

NEXT STEPS

Ready to Build & Deploy on GitHub

A high-impact, realistic web application addressing genuine
healthcare logistical bottlenecks.

 Repo Setup  Fast MVP  Demo Ready

Image Sources



https://photos.peopleimages.com/picture/202208/2481539-doctor-consulting-patient-with-digital-tablet-giving-diagnosis-and-medical-checkup-in-a-hospital.-healthcare-worker-and-trusted-physician-in-appointment-with-medicine-advice-wellness-and-feedback-fit_400_400.jpg

Source: peopleimages.com